

21bca071

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Sem-4

prg 1:

write a program to evaluate simple interest of a given principal rate and time

class main

{

public static void main(String args[])

{

float p,r,si,t;

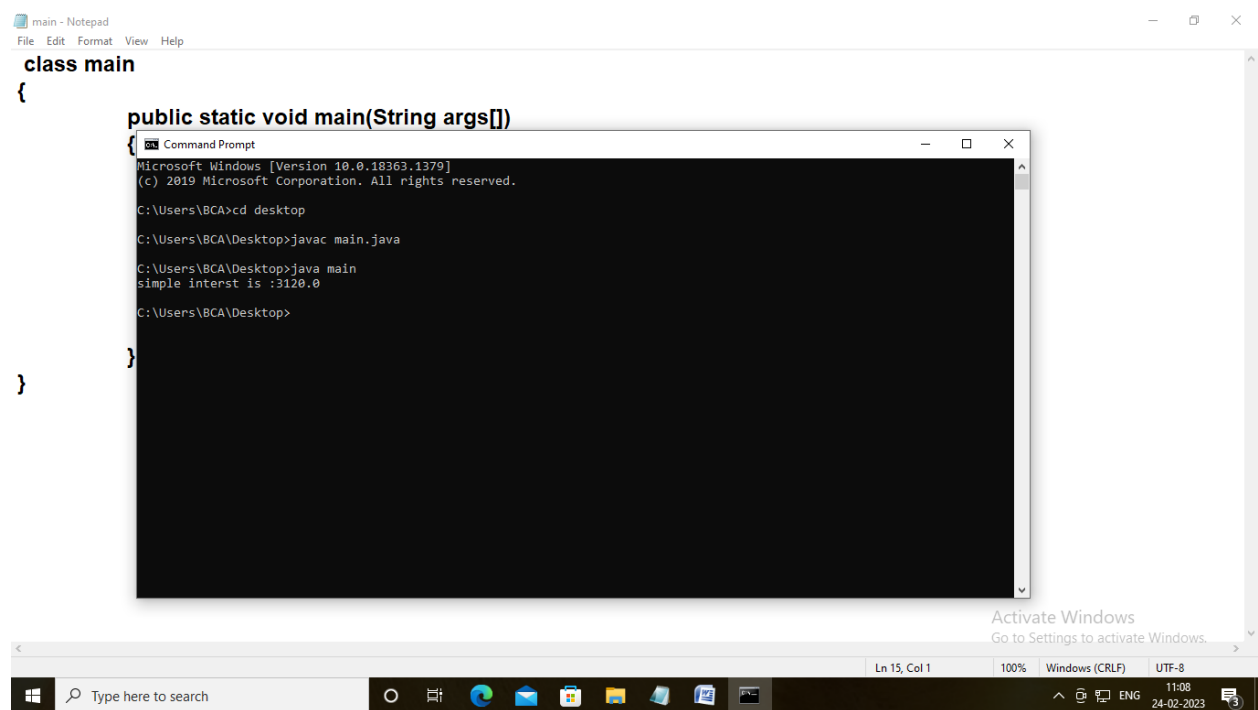
// principal amount time , rate and simple
interest respectively

p=13000; r=12;t=2;

si=(p*r*t)/100;

System.out.println("simple interest is :"+si);

```
}  
  
}  
  
main - Notepad  
File Edit Format View Help  
class main  
{  
    public static void main(String args[])  
    {  
        Command Prompt  
        Microsoft Windows [Version 10.0.18363.1370]  
        (c) 2019 Microsoft Corporation. All rights reserved.  
  
        C:\Users\BCA>cd desktop  
  
        C:\Users\BCA\Desktop>javac main.java  
  
        C:\Users\BCA\Desktop>java main  
        simple interest is :3120.0  
  
        C:\Users\BCA\Desktop>  
    }  
}  
}
```



prg 2:

a motor cycle dealer sells two wheelers to this customer on loan, which is to be repaid in 5 years. the dealer charges simple interest for the whole term on the day of giving the loan itself. the total amount is then divided by 60(month) and is collected as equated monthly instalment(emi). write program to calculate the emi for a

loan of rs .x where x is givem from command line argument . print the emi value in rupees.

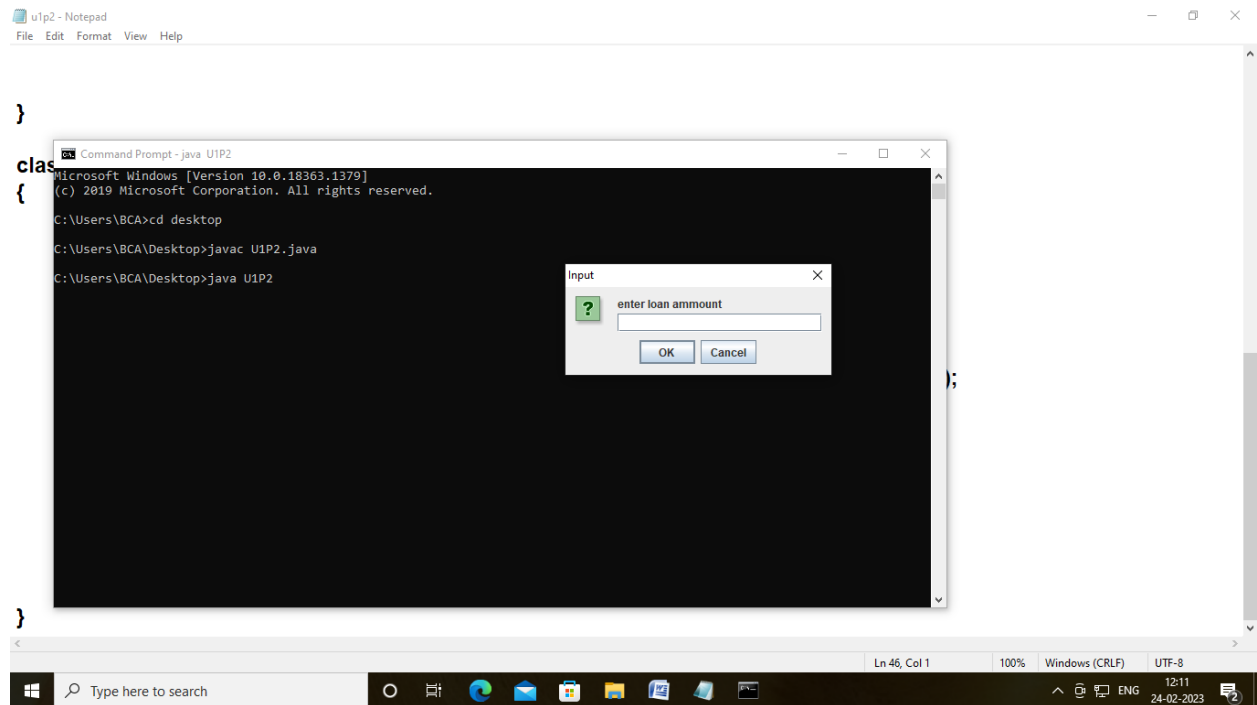
```
import javax.swing.JOptionPane;

class loan
{
    int loan;
    int interest;
    double emi;

    loan(int loan , int interest)
    {
        this.loan=loan;
        this.interest=interest;
    }

    double calcemi()
    {
        return loan * interest/(100*60);
    }
}
```

```
}  
  
class u1p2  
{  
    public static void main(String arg[])  
    {  
        int loan;  
        int interest;  
        String  
lo=JOptionPane.showInputDialog(null,"enter loan  
ammount");  
        loan=Integer.parseInt(lo);  
        String  
in=JOptionPane.showInputDialog(null,"enter interest ");  
        interest=Integer.parseInt(in);  
        loan l1=new loan(loan,interest);  
        System.out.println("emi="+l1.calcemi());  
    }  
}
```



prg 3:

a car accessories shop assigns code 1 to seat covers , 2 to steering wheel covers, 3 to cae lighting and 4 for air purifiers . all other items have code 5 or more. while selling the goods, a sales tax of 2% to seat covers , 3% to steering wheel covers , 4% to car lighting , 2.5% to air purifiers and 1.2% for all other items is charged. a list containing the product code and price is given for making a bill.write a java program using switch statements to prepare a bill.

```
import javax.swing.JOptionPane;

class CarAcc
{
    static int prodPrice[][]={{1,100},
        {2,200},
        {3,300},
        {4,400},
        {5,500},
        {6,600}};

    static double salesTax[]={2,3,42.5,1.2};

    CarAcc()
    {}

    void getBill(int prCode,int qty)
    {
        double billAmm;

        switch(prCode)
        {
```

case 1:

```
        billAmm=(prodPrice[prCode-1][1]*qty*salesTax[prCode-1]/100)+prodPrice[prCode-1][1]*qty;
```

```
        JOptionPane.showMessageDialog(null,"price="+billAmm);
```

```
        break;
```

case 2:

```
        billAmm=(prodPrice[prCode-1][1]*qty*salesTax[prCode-1]/100)+prodPrice[prCode-1][1]*qty;
```

```
        JOptionPane.showMessageDialog(null,"Price="+billAmm);
```

```
        break;
```

case 3:

```
        billAmm=(prodPrice[prCode-1][1]*qty*salesTax[prCode-1]/100)+prodPrice[prCode-1][1]*qty;
```

```
        JOptionPane.showMessageDialog(null,"Price="+billAmm);
```

```
break;
```

```
case 4:
```

```
        billAmm=(prodPrice[prCode-1][1]*qty*salesTax[prCode-1]/100)+prodPrice[prCode-1][1]*qty;
```

```
        JOptionPane.showMessageDialog(null,"Price="+billAmm);
```

```
break;
```

```
case 5:
```

```
        billAmm=(prodPrice[prCode-1][1]*qty*salesTax[prCode-1]/100)+prodPrice[prCode-1][1]*qty;
```

```
        JOptionPane.showMessageDialog(null,"Price="+billAmm);
```

```
break;
```

```
default:System.out.println("Please enter proper value");
```

```
}
```

```

    }
}
class U1P3
{
    public static void main(String arg[])
    {
        int prCode,qty;

        CarAcc b1=new CarAcc();

        String
a=JOptionPane.showInputDialog(null,"enter prod
Code:");

        prCode=Integer.parseInt(a);

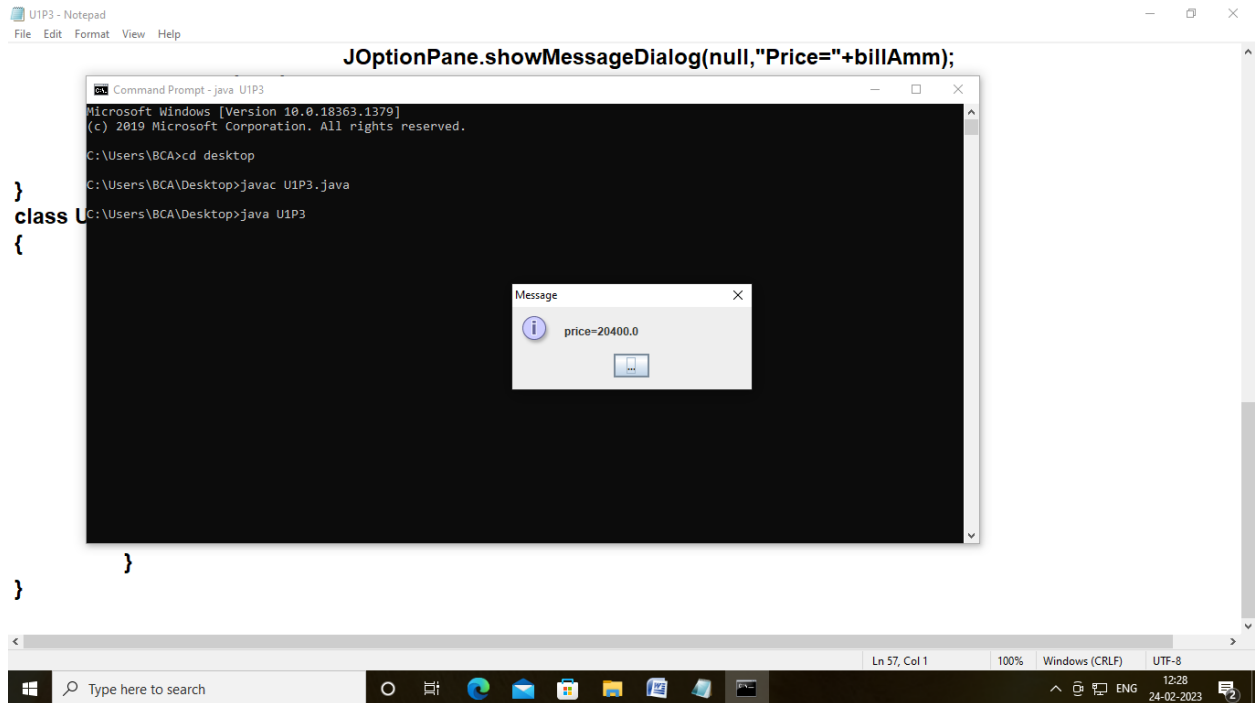
        String
b=JOptionPane.showInputDialog(null,"enter Quantity:");

        qty=Integer.parseInt(b);

        b1.getBill(prCode,qty);

    }
}

```



prg 4:

write a java program to scan 3 integer value from the command line arugument and display the maximum number using a conditional operator .

class u1p

{

public static void main(String arg[])

{

```
int a,b,c,max;
```

```
    a=Integer.parseInt(arg[0]);
```

```
b=Integer.parseInt(arg[1]);
```

```
    c=Integer.parseInt(arg[2]);
```

```
    if(a>b)
```

```
    {
```

```
        if(a>c)
```

```
        max=a;
```

```
    else
```

```
        max=c;
```

```
    else
```

```
    {
```

```
        if(b>c)
```

```
        max=b;
```

```
    else
```

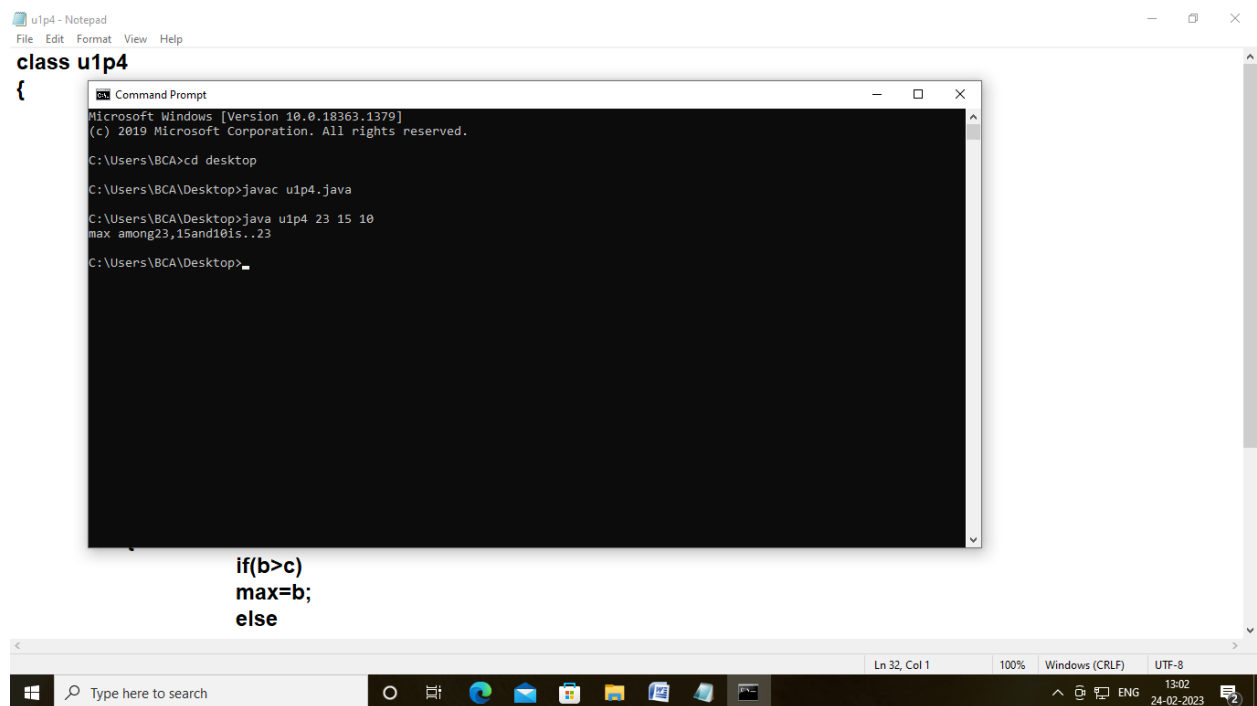
```
        max =c;
```

```
}
```

```
System.out.println("max  
among"+a+", "+b+"and "+c+"is.." +max);
```

```
}
```

```
}
```



```
u1p4 - Notepad
File Edit Format View Help
class u1p4
{
    Command Prompt
    Microsoft Windows [Version 10.0.18363.1379]
    (c) 2019 Microsoft Corporation. All rights reserved.

    C:\Users\BCA>cd desktop
    C:\Users\BCA\Desktop>javac u1p4.java
    C:\Users\BCA\Desktop>java u1p4 23 15 10
    max among 23, 15 and 10 is .. 23
    C:\Users\BCA\Desktop>

    if(b>c)
    max=b;
    else
```

prg 5:

write a java program to calculate the hypotenuse of right angled triangle when other sides of the triangle are

giiven.(hypotenuse=square root ($x^2 + y^2$) using
command line argument.

```
class u1p5
```

```
{
```

```
    public static void main(String arg[])
```

```
    {
```

```
        int x,y,z;
```

```
        double hyp;
```

```
        x=Integer.parseInt(arg[0]);
```

```
        y=Integer.parseInt(arg[1]);
```

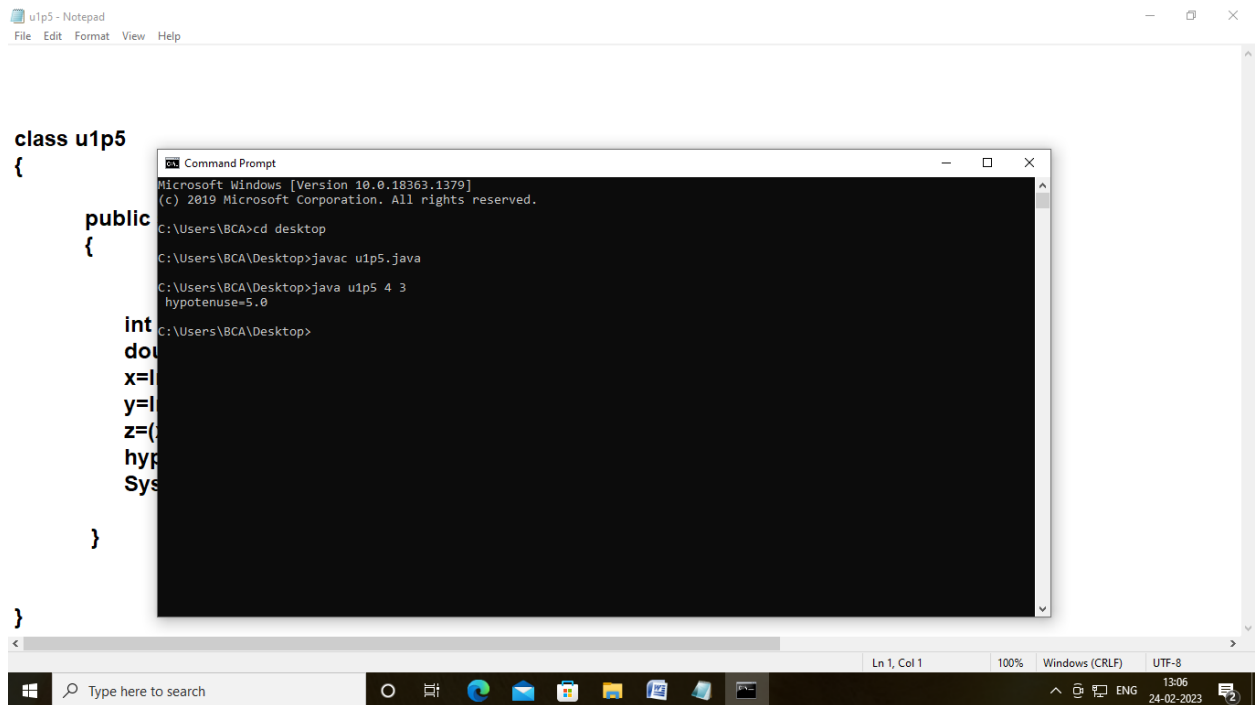
```
        z=(x*x)+(y*y);
```

```
        hyp=Math.sqrt(z);
```

```
        System.out.println(" hypotenuse="+hyp);
```

```
    }
```

```
}
```



prg 6:

write a java program to calculate the area of square and rectangle by overloading the area method.

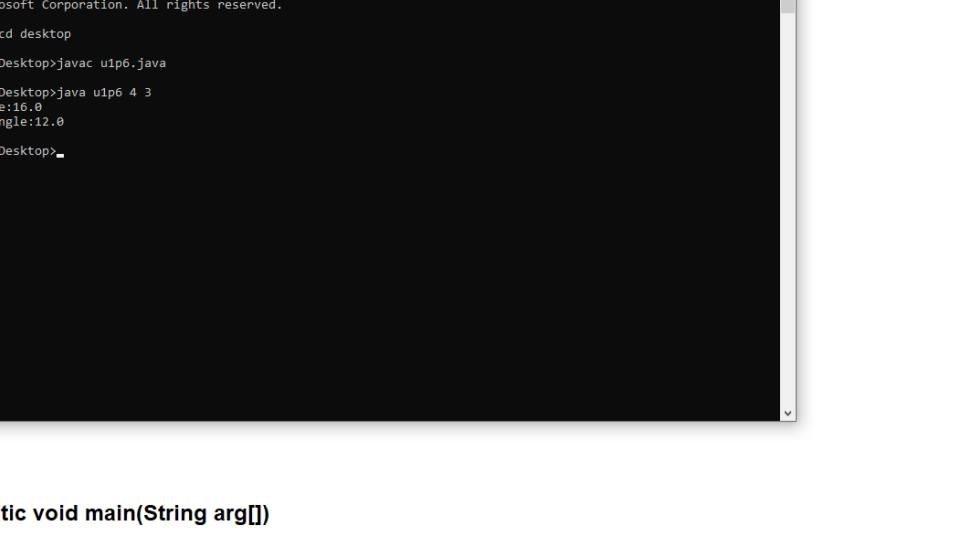
```
class areacalc
```

```
{
```

```
    double area(int x)
```

```
{
```

```
        return x*x;
    }
    double area(int x , int y)
    {
        return x*y;
    }
}
class u1p6
{
    public static void main(String arg[])
    {
        int x,y;
        x=Integer.parseInt(arg[0]);
        y=Integer.parseInt(arg[1]);
        areacalc a=new areacalc();
        System.out.println("area of square:"+a.area(x));
        System.out.println("area of rectangle:"+a.area(x,y));
    }
}
```



The screenshot displays a Windows desktop environment. In the foreground, a Notepad window titled 'u1p6 - Notepad' is open, showing the following Java code:

```
class u1p6
{
    public static void main(String arg[])
    {
```

Below the Notepad window, a Command Prompt window titled 'Command Prompt' is open, showing the execution of the code. The output is as follows:

```
C:\Users\BCA>cd desktop
C:\Users\BCA\Desktop>javac u1p6.java
C:\Users\BCA\Desktop>java u1p6 4 3
area of square:16.0
area of rectangle:12.0
C:\Users\BCA\Desktop>
```

The taskbar at the bottom of the screen shows the Start button, a search bar, and several pinned applications including Edge, Mail, File Explorer, and the Task Manager. The system tray on the right indicates the date and time as 13:10 on 24-02-2023.

create a complex number class. the class should have a constructor and method to add, subtract and multiple

two complex number and to return the real and imaginary parts.

```
class complex
```

```
{
```

```
    int real,image;
```

```
    complex()
```

```
}
```

```
    complex(int real1,int image1)
```

```
{
```

```
    real =real1;
```

```
    image=image1;
```

```
}
```

```
complex sumcomplex(complex c1,complex c2)
```

```
{  
    complex csum=new complex();  
    csum.real=c1.real + c2.real;  
    csum.image=c1.image+ c2.image;  
    return csum;  
}
```

```
complex subcomplex(complex c1,complex c2)
```

```
{  
    complex csub=new complex();  
    csub.real=c1.real - c2.real;  
    csub.image=c1.image - c2.image;  
    return csub;  
}
```

```
complex mulcomplex(complex c1,complex c2)
```

```
{
```

```
    complex cmul=new complex();  
    cmul.real=c1.real * c2.real - c1.image*c2.image;  
    cmul.image =c1.real*c2.image+c1.image*c2.real;  
    return cmul;  
}  
  
}
```

```
class u1p7
```

```
{
```

```
    public static void main(String []a)
```

```
{
```

```
        complex c1=new complex(4,8);
```

```
        complex c2=new complex(5,7);
```

```
        complex c3=new complex();
```

```
c3=c3.csumcomplex(c1,c2);
```

```
System.out.println("sum"+c3.real+"+i"+c3.image);
```

```
c3=c3.subcomplex(c1,c2);
```

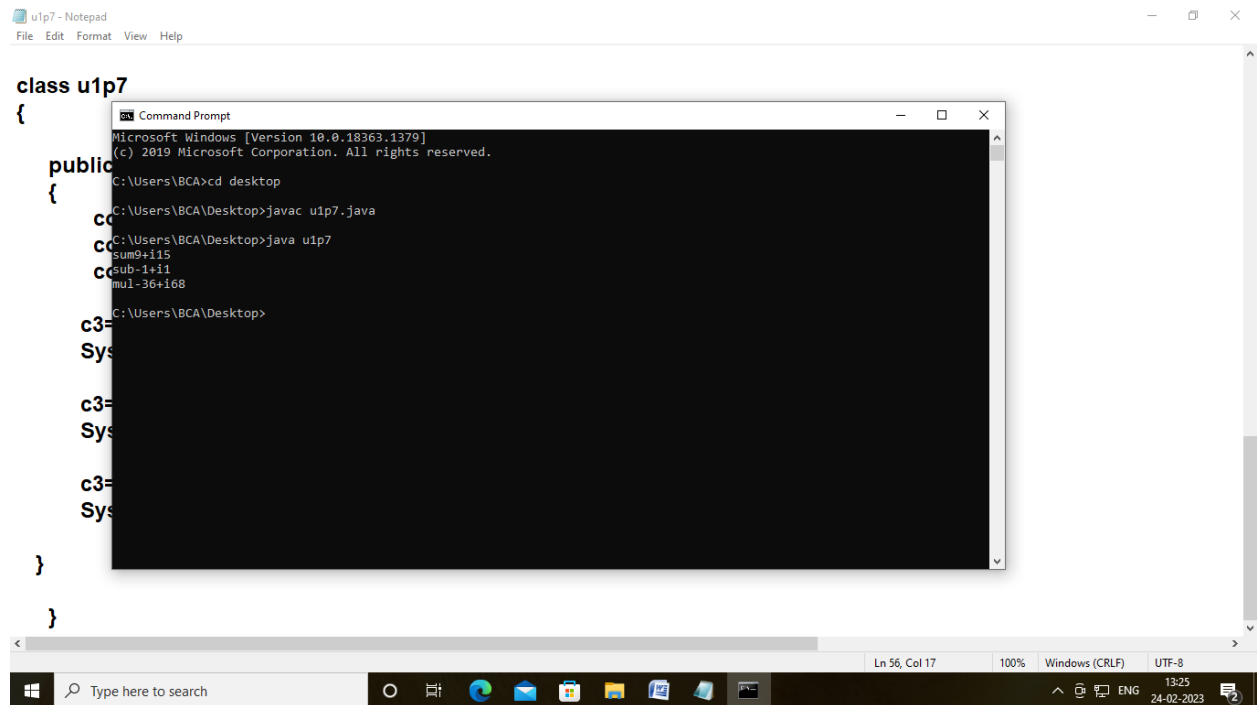
```
System.out.println("sub"+c3.real+"+i"+c3.image);
```

```
c3=c3.mulcomplex(c1,c2);
```

```
System.out.println("mul"+c3.real+"+i"+c3.image);
```

```
}
```

```
}
```



prg 8:

write a java program to display power of 2 i.e 2,4,8,16 ,
etc up to 1024 using bitwise operators.

```
class u1p8
```

```
{
```

```
    public static void main(String arg[])
```

```
{  
  
    int a=1;  
    for(int i=1;i<=10;i++)  
    {  
  
        a=a<<1;  
        System.out.println(a);  
  
    }  
  
}  
  
}
```

